

1. Supplemental Table: Neutral Reactions

The neutral chemical reactions used in our models are collected in Supplemental Table S.1:

Table S.1. Neutral chemical reactions in the model

Index	Reaction	Rate Coefficient	Reference
R1	$\text{CO} + \text{NO}_3 \rightarrow \text{NO}_2 + \text{CO}_2$	4.00×10^{-19}	(1)
R2	$\text{CO} + \text{S}_3 \rightarrow \text{COS} + \text{S}_2$	$1.00 \times 10^{-11} \exp(-20000.0/T)$	(2)
R3	$\text{CO} + \text{SH} \rightarrow \text{H} + \text{COS}$	$6.30 \times 10^{-14} \exp(-7780.0/T)$	(2)
R4	$\text{CO} + \text{SO}_2 \rightarrow \text{CO}_2 + \text{SO}$	$4.50 \times 10^{-12} \exp(-24300.0/T)$	(3)
R5	$\text{COCl} + \text{CO}_2 \rightarrow \text{CO} + \text{Cl} + \text{CO}_2$	$k_{eq}=1.16 \times 10^{-24} \exp(3730.0/T)$	(4)
R6	$\text{COCl} + \text{COCl} \rightarrow \text{COCl}_2 + \text{CO}$	5.00×10^{-11}	(3)
R7	$\text{COCl} + \text{Cl}_2 \rightarrow \text{COCl}_2 + \text{Cl}$	$6.45 \times 10^{-12} \times (1.03 \times 10^{21}/T + n_{\text{atm}}) \times k(\text{COCl} + \text{O}_2 + \text{M})$	(3)
R8	$\text{COCl} + \text{ClCO}_3 \rightarrow \text{CO}_2 + \text{CO}_2 + \text{Cl} + \text{Cl}$	1.00×10^{-11}	(3)
R9	$\text{COCl} + \text{N}_2 \rightarrow \text{CO} + \text{Cl} + \text{N}_2$	$k_{eq}=3.50 \times 10^{-25} \exp(3730.0/T)$	(4)
R10	$\text{COCl} + \text{NO}_2 \rightarrow \text{CO}_2 + \text{NO} + \text{Cl}$	$6.00 \times 10^{-13} \exp(600.0/T)$	(5)
R11	$\text{COS} + \text{M} \rightarrow \text{CO} + \text{S} + \text{M}$	$2.20 \times 10^{-07} \exp(-37300.0/T)$	(6)
R12	$\text{COS} + \text{NO}_3 \rightarrow \text{CO} + \text{SO} + \text{NO}_2$	1.00×10^{-16}	(1)
R13	$\text{COS} + \text{S}_2 \rightarrow \text{CO} + \text{S}_3$	$2.50 \times 10^{-11} \exp(-25500.0/T)$	(2)
R14	$\text{COS} + \text{SCl} \rightarrow \text{ClS}_2 + \text{CO}$	3.00×10^{-16}	(3)
R15	$\text{Cl} + \text{COCl} \rightarrow \text{Cl}_2 + \text{CO}$	$2.16 \times 10^{-09} \exp(-1670.0/T)$	(7)
R16	$\text{Cl} + \text{COS} \rightarrow \text{SCl} + \text{CO}$	1.00×10^{-16}	(1)
R17	$\text{Cl} + \text{ClCO}_3 \rightarrow \text{Cl} + \text{ClO} + \text{CO}_2$	1.00×10^{-11}	(8)
R18	$\text{Cl} + \text{ClS}_2 \rightarrow \text{Cl}_2 + \text{S}_2$	1.00×10^{-11}	(9)
R19	$\text{Cl} + \text{ClSO}_2 \rightarrow \text{SO}_2 + \text{Cl}_2$	1.00×10^{-20}	(10)
R20	$\text{Cl} + \text{H}_2 \rightarrow \text{HCl} + \text{H}$	$3.05 \times 10^{-11} \exp(-2270.0/T)$	(1)
R21	$\text{Cl} + \text{H}_2\text{O} \rightarrow \text{OH} + \text{HCl}$	$2.00 \times 10^{-11} \exp(-8470.0/T)$	(2)
R22	$\text{Cl} + \text{H}_2\text{O}_2 \rightarrow \text{HCl} + \text{HO}_2$	$1.10 \times 10^{-11} \exp(-980.0/T)$	(1)
R23	$\text{Cl} + \text{H}_2\text{S} \rightarrow \text{SH} + \text{HCl}$	$3.70 \times 10^{-11} \exp(210.0/T)$	(11)
R24	$\text{Cl} + \text{HNO}_3 \rightarrow \text{HCl} + \text{NO}_3$	2.00×10^{-16}	(1)
R25	$\text{Cl} + \text{HO}_2 \rightarrow \text{HCl} + \text{O}_2$	$1.40 \times 10^{-11} \exp(270.0/T)$	(1)
R26	$\text{Cl} + \text{HO}_2 \rightarrow \text{OH} + \text{ClO}$	$3.60 \times 10^{-11} \exp(-375.0/T)$	(1)
R27	$\text{Cl} + \text{HOCl} \rightarrow \text{HCl} + \text{ClO}$	$5.89 \times 10^{-25} T^{4.070} \exp(169.8/T)$	(12)
R28	$\text{Cl} + \text{HOCl} \rightarrow \text{OH} + \text{Cl}_2$	$3.40 \times 10^{-12} \exp(-130.0/T)$	(1)
R29	$\text{Cl} + \text{NO}_3 \rightarrow \text{ClO} + \text{NO}_2$	2.40×10^{-11}	(1)
R30	$\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$	$2.30 \times 10^{-11} \exp(-200.0/T)$	(1)
R31	$\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2(^1\Delta)$	$5.80 \times 10^{-13} \exp(-260.0/T)$	(4)
R32	$\text{Cl} + \text{OH} \rightarrow \text{HCl} + \text{O}$	$9.80 \times 10^{-12} \exp(-2860.0/T)$	(7)
R33	$\text{Cl} + \text{OSCl} \rightarrow \text{Cl}_2 + \text{SO}$	2.30×10^{-11}	(9)
R34	$\text{Cl} + \text{S}_2\text{Cl}_2 \rightarrow \text{Cl}_2 + \text{ClS}_2$	4.30×10^{-12}	(3)
R35	$\text{Cl} + \text{S}_4 \rightarrow \text{ClS}_2 + \text{S}_2$	2.00×10^{-12}	(9)
R36	$\text{Cl} + \text{SCl} \rightarrow \text{S} + \text{Cl}_2$	1.00×10^{-14}	(3)
R37	$\text{Cl} + \text{SCl}_2 \rightarrow \text{Cl}_2 + \text{SCl}$	1.00×10^{-19}	(3)
R38	$\text{Cl} + \text{SH} \rightarrow \text{HCl} + \text{S}$	$8.00 \times 10^{-11} \exp(210.0/T)$	(13)
R39	$\text{Cl} + \text{SO}_2\text{Cl}_2 \rightarrow \text{ClSO}_2 + \text{Cl}_2$	$1.00 \times 10^{-11} \exp(-1000.0/T)$	(2)
R40	$\text{ClCO}_3 + \text{ClCO}_3 \rightarrow \text{Cl} + \text{Cl} + \text{CO}_2 + \text{CO}_2 + \text{O}_2$	5.00×10^{-12}	(3)
R41	$\text{ClCO}_3 + \text{SO} \rightarrow \text{Cl} + \text{SO}_2 + \text{CO}_2$	1.00×10^{-11}	(3)
R42	$\text{ClCO}_3 + \text{SO}_2 \rightarrow \text{Cl} + \text{SO}_3 + \text{CO}_2$	1.00×10^{-15}	(3)
R43	$\text{ClO} + \text{CO} \rightarrow \text{CO}_2 + \text{Cl}$	$1.00 \times 10^{-12} \exp(-3700.0/T)$	(1)
R44	$\text{ClO} + \text{COS} \rightarrow \text{OSCl} + \text{CO}$	2.00×10^{-16}	(1)
R45	$\text{ClO} + \text{ClO} \rightarrow \text{Cl}_2 + \text{O}_2$	$1.00 \times 10^{-12} \exp(-1590.0/T)$	(4)
R46	$\text{ClO} + \text{ClO} \rightarrow \text{Cl}_2 + \text{O}_2(^1\Delta)$	$2.00 \times 10^{-14} \exp(-1590.0/T)$	(4)
R47	$\text{ClO} + \text{H}_2 \rightarrow \text{HCl} + \text{OH}$	$1.00 \times 10^{-12} \exp(-4800.0/T)$	(1)
R48	$\text{ClO} + \text{HO}_2 \rightarrow \text{HOCl} + \text{O}_2$	$2.60 \times 10^{-12} \exp(290.0/T)$	(1)
R49	$\text{ClO} + \text{N}_2\text{O} \rightarrow \text{NO} + \text{NO} + \text{Cl}$	$1.00 \times 10^{-12} \exp(-4300.0/T)$	(1)
R50	$\text{ClO} + \text{NO} \rightarrow \text{Cl} + \text{NO}_2$	$6.40 \times 10^{-12} \exp(290.0/T)$	(1)
R51	$\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2$	$2.80 \times 10^{-11} \exp(85.0/T)$	(1)
R52	$\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2(^1\Delta)$	$1.40 \times 10^{-14} \exp(85.0/T)$	(1)
R53	$\text{ClO} + \text{OH} \rightarrow \text{HCl} + \text{O}_2$	$6.00 \times 10^{-13} \exp(230.0/T)$	(1)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
R54	$\text{ClO} + \text{OH} \rightarrow \text{HO}_2 + \text{Cl}$	$7.40 \times 10^{-12} \exp(270.0/T)$	(1)
R55	$\text{ClO} + \text{S}_2 \rightarrow \text{S}_2\text{O} + \text{Cl}$	2.80×10^{-11}	(10)
R56	$\text{ClO} + \text{SO} \rightarrow \text{Cl} + \text{SO}_2$	2.80×10^{-11}	(1)
R57	$\text{ClO} + \text{SO}_2 \rightarrow \text{Cl} + \text{SO}_3$	4.00×10^{-18}	(1)
R58	$\text{ClSO}_2 + \text{ClSO}_2 \rightarrow \text{Cl}_2 + \text{SO}_2 + \text{SO}_2$	5.00×10^{-13}	(10)
R59	$\text{ClSO}_2 + \text{ClSO}_2 \rightarrow \text{SO}_2\text{Cl}_2 + \text{SO}_2$	5.00×10^{-11}	(2)
R60	$\text{ClSO}_2 + \text{M} \rightarrow \text{Cl} + \text{SO}_2 + \text{M}$	$7.00 \times 10^{-16} \exp(-10540.0/T)$	(2)
R61	$\text{H} + \text{CO}_2 \rightarrow \text{CO} + \text{OH}$	$1.00 \times 10^{-10} \exp(-12400.0/T)$	(6)
R62	$\text{H} + \text{COCl} \rightarrow \text{HCl} + \text{CO}$	1.00×10^{-11}	(8)
R63	$\text{H} + \text{COS} \rightarrow \text{CO} + \text{SH}$	$1.20 \times 10^{-11} \exp(-1950.0/T)$	(14)
R64	$\text{H} + \text{Cl}_2 \rightarrow \text{HCl} + \text{Cl}$	$1.43 \times 10^{-10} \exp(-590.0/T)$	(3)
R65	$\text{H} + \text{ClCO}_3 \rightarrow \text{Cl} + \text{OH} + \text{CO}_2$	1.00×10^{-11}	(8)
R66	$\text{H} + \text{ClSO}_2 \rightarrow \text{SO}_2 + \text{HCl}$	1.00×10^{-11}	(3)
R67	$\text{H} + \text{H}_2\text{S} \rightarrow \text{SH} + \text{H}_2$	$3.71 \times 10^{-12} T^{1.970} \exp(-455.0/T)$	(14)
R68	$\text{H} + \text{HCl} \rightarrow \text{H}_2 + \text{Cl}$	$1.50 \times 10^{-11} \exp(-1750.0/T)$	(3)
R69	$\text{H} + \text{HNO} \rightarrow \text{H}_2 + \text{NO}$	1.30×10^{-10}	(15)
R70	$\text{H} + \text{HO}_2 \rightarrow \text{H}_2 + \text{O}_2$	6.90×10^{-12}	(1)
R71	$\text{H} + \text{HO}_2 \rightarrow \text{H}_2 + \text{O}_2(^1\Delta)$	1.30×10^{-13}	(3)
R72	$\text{H} + \text{HO}_2 \rightarrow \text{H}_2\text{O} + \text{O}$	1.60×10^{-12}	(1)
R73	$\text{H} + \text{HO}_2 \rightarrow \text{OH} + \text{OH}$	7.20×10^{-11}	(1)
R74	$\text{H} + \text{NO}_2 \rightarrow \text{OH} + \text{NO}$	1.35×10^{-10}	(1)
R75	$\text{H} + \text{NO}_3 \rightarrow \text{OH} + \text{NO}_2$	1.10×10^{-10}	(16)
R76	$\text{H} + \text{O}_3 \rightarrow \text{OH} + \text{O}_2$	$1.40 \times 10^{-10} \exp(-470.0/T)$	(1)
R77	$\text{H} + \text{S}_3 \rightarrow \text{SH} + \text{S}_2$	$1.20 \times 10^{-10} \exp(-1950.0/T)$	(2)
R78	$\text{H} + \text{SCl} \rightarrow \text{HCl} + \text{S}$	1.00×10^{-11}	(2)
R79	$\text{H} + \text{SCl}_2 \rightarrow \text{HCl} + \text{SCl}$	2.00×10^{-11}	(3)
R80	$\text{H} + \text{SH} \rightarrow \text{H}_2 + \text{S}$	2.16×10^{-11}	(14)
R81	$\text{H} + \text{SNO} \rightarrow \text{NO} + \text{SH}$	$4.00 \times 10^{-10} \exp(-340.0/T)$	(2)
R82	$\text{H}_2\text{O} + \text{H} \rightarrow \text{OH} + \text{H}_2$	$1.30 \times 10^{-11} \exp(-9420.0/T)$	(2)
R83	$\text{H}_2\text{S} + \text{NO}_3 \rightarrow \text{SH} + \text{HNO}_3$	8.00×10^{-16}	(1)
R84	$\text{H}_2\text{S} + \text{O}_3 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$	2.00×10^{-20}	(1)
R85	$\text{H}_2\text{SO}_4 + \text{H}_2\text{O} \rightarrow \text{SO}_3 + \text{H}_2\text{O} + \text{H}_2\text{O}$	$7.00 \times 10^{-14} \exp(-5170.0/T)$	(4)
R86	$\text{HCl} + \text{NO}_3 \rightarrow \text{HNO}_3 + \text{Cl}$	5.00×10^{-17}	(1)
R87	$\text{HNO} + \text{O}_2 \rightarrow \text{NO} + \text{HO}_2$	$5.25 \times 10^{-12} \exp(-1510.0/T)$	(17)
R88	$\text{HO}_2 + \text{HO}_2 \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$	$3.00 \times 10^{-13} \exp(460.0/T)$	(1)
R89	$\text{HO}_2 + \text{HO}_2 \rightarrow \text{H}_2\text{O}_2 + \text{O}_2(^1\Delta)$	$4.60 \times 10^{-15} \exp(460.0/T)$	(15)
R90	$\text{HO}_2 + \text{NO} \rightarrow \text{NO}_2 + \text{OH}$	$3.44 \times 10^{-12} \exp(260.0/T)$	(1)
R91	$\text{HO}_2 + \text{NO}_2 \rightarrow \text{HNO}_2 + \text{O}_2$	5.00×10^{-16}	(1)
R92	$\text{HO}_2 + \text{NO}_3 \rightarrow \text{HNO}_3 + \text{O}_2$	3.50×10^{-12}	(1)
R93	$\text{HO}_2 + \text{O}_3 \rightarrow \text{OH} + \text{O}_2 + \text{O}_2$	$1.00 \times 10^{-14} \exp(-490.0/T)$	(1)
R94	$\text{HO}_2 + \text{SO} \rightarrow \text{SO}_2 + \text{OH}$	2.80×10^{-11}	(15)
R95	$\text{HO}_2 + \text{SO}_2 \rightarrow \text{OH} + \text{SO}_3$	1.00×10^{-18}	(1)
R96	$\text{HSCl} + \text{Cl} \rightarrow \text{HCl} + \text{SCl}$	$2.50 \times 10^{-13} \exp(-130.0/T)$	(2)
R97	$\text{HSCl} + \text{H} \rightarrow \text{H}_2 + \text{SCl}$	$1.20 \times 10^{-13} \exp(-2770.0/T)$	(2)
R98	$\text{HSCl} + \text{S} \rightarrow \text{SH} + \text{SCl}$	1.70×10^{-13}	(2)
R99	$\text{HSCl} + \text{SH} \rightarrow \text{H}_2\text{S} + \text{SCl}$	$3.00 \times 10^{-12} \exp(-500.0/T)$	(2)
R100	$\text{HSO}_3 + \text{O}_2 \rightarrow \text{HO}_2 + \text{SO}_3$	$1.30 \times 10^{-12} \exp(-330.0/T)$	(4)
R101	$\text{N} + \text{HO}_2 \rightarrow \text{NO} + \text{OH}$	2.20×10^{-11}	(18)
R102	$\text{N} + \text{NO} \rightarrow \text{N}_2 + \text{O}$	$2.10 \times 10^{-11} \exp(100.0/T)$	(1)
R103	$\text{N} + \text{NO}_2 \rightarrow \text{N}_2\text{O} + \text{O}$	$5.80 \times 10^{-12} \exp(220.0/T)$	(1)
R104	$\text{N} + \text{O}_2 \rightarrow \text{NO} + \text{O}$	$3.30 \times 10^{-12} \exp(-3150.0/T)$	(1)
R105	$\text{N} + \text{O}_3 \rightarrow \text{NO} + \text{O}_2$	2.00×10^{-16}	(1)
R106	$\text{N} + \text{OH} \rightarrow \text{NO} + \text{H}$	$3.80 \times 10^{-11} \exp(85.0/T)$	(19)
R107	$\text{NO} + \text{NO}_3 \rightarrow \text{NO}_2 + \text{NO}_2$	$1.70 \times 10^{-11} \exp(125.0/T)$	(1)
R108	$\text{NO}_3 + \text{NO}_2 \rightarrow \text{NO} + \text{NO}_2 + \text{O}_2$	$4.35 \times 10^{-14} \exp(-1335.0/T)$	(1)
R109	$\text{NO}_3 + \text{NO}_3 \rightarrow \text{NO}_2 + \text{NO}_2 + \text{O}_2$	$8.50 \times 10^{-13} \exp(-2450.0/T)$	(1)
R110	$\text{O} + \text{COCl} \rightarrow \text{CO} + \text{ClO}$	3.00×10^{-12}	(8)
R111	$\text{O} + \text{COCl} \rightarrow \text{Cl} + \text{CO}_2$	3.00×10^{-11}	(8)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
R112	$\text{O} + \text{COS} \rightarrow \text{SO} + \text{CO}$	$2.10 \times 10^{-11} \exp(-2200.0/T)$	(20)
R113	$\text{O} + \text{Cl}_2 \rightarrow \text{ClO} + \text{Cl}$	$7.40 \times 10^{-12} \exp(-1650.0/T)$	(3)
R114	$\text{O} + \text{ClCO}_3 \rightarrow \text{Cl} + \text{O}_2 + \text{CO}_2$	1.00×10^{-11}	(8)
R115	$\text{O} + \text{ClS}_2 \rightarrow \text{SO} + \text{SCl}$	1.00×10^{-13}	(10)
R116	$\text{O} + \text{ClSO}_2 \rightarrow \text{SO}_2 + \text{ClO}$	1.00×10^{-11}	(3)
R117	$\text{O} + \text{H}_2 \rightarrow \text{OH} + \text{H}$	$8.50 \times 10^{-20} T^{2.670} \exp(-3160.0/T)$	(21)
R118	$\text{O} + \text{H}_2\text{O}_2 \rightarrow \text{OH} + \text{HO}_2$	$1.40 \times 10^{-12} \exp(-2000.0/T)$	(1)
R119	$\text{O} + \text{H}_2\text{S} \rightarrow \text{OH} + \text{SH}$	$9.20 \times 10^{-12} \exp(-1800.0/T)$	(1)
R120	$\text{O} + \text{HCl} \rightarrow \text{OH} + \text{Cl}$	$1.00 \times 10^{-11} \exp(-3300.0/T)$	(1)
R121	$\text{O} + \text{HNO}_3 \rightarrow \text{OH} + \text{NO}_3$	3.00×10^{-17}	(1)
R122	$\text{O} + \text{HO}_2 \rightarrow \text{OH} + \text{O}_2$	$3.00 \times 10^{-11} \exp(200.0/T)$	(1)
R123	$\text{O} + \text{HO}_2 \rightarrow \text{OH} + \text{O}_2(^1\Delta)$	$6.00 \times 10^{-13} \exp(200.0/T)$	(3)
R124	$\text{O} + \text{HOCl} \rightarrow \text{OH} + \text{ClO}$	1.70×10^{-13}	(1)
R125	$\text{O} + \text{NO}_2 \rightarrow \text{NO} + \text{O}_2$	$5.30 \times 10^{-12} \exp(200.0/T)$	(1)
R126	$\text{O} + \text{NO}_3 \rightarrow \text{O}_2 + \text{NO}_2$	1.30×10^{-11}	(1)
R127	$\text{O} + \text{O}_3 \rightarrow \text{O}_2 + \text{O}_2$	$8.00 \times 10^{-12} \exp(-2060.0/T)$	(1)
R128	$\text{O} + \text{OH} \rightarrow \text{O}_2 + \text{H}$	$1.80 \times 10^{-11} \exp(180.0/T)$	(1)
R129	$\text{O} + \text{OSCl} \rightarrow \text{SO} + \text{ClO}$	$2.00 \times 10^{-11} \exp(-600.0/T)$	(10)
R130	$\text{O} + \text{OSCl} \rightarrow \text{SO}_2 + \text{Cl}$	$5.00 \times 10^{-11} \exp(-600.0/T)$	(10)
R131	$\text{O} + \text{S}_2 \rightarrow \text{SO} + \text{S}$	$2.20 \times 10^{-11} \exp(-84.0/T)$	(10)
R132	$\text{O} + \text{S}_2\text{O} \rightarrow \text{SO} + \text{SO}$	1.70×10^{-12}	(3)
R133	$\text{O} + \text{S}_2\text{O}_2 \rightarrow \text{S}_2\text{O} + \text{O}_2$	3.00×10^{-14}	(3)
R134	$\text{O} + \text{S}_2\text{O}_2 \rightarrow \text{SO} + \text{SO}_2$	3.00×10^{-15}	(3)
R135	$\text{O} + \text{S}_3 \rightarrow \text{SO} + \text{S}_2$	8.00×10^{-11}	(10)
R136	$\text{O} + \text{S}_4 \rightarrow \text{SO} + \text{S}_3$	8.00×10^{-11}	(10)
R137	$\text{O} + \text{S}_5 \rightarrow \text{S}_4 + \text{SO}$	$8.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R138	$\text{O} + \text{S}_6 \rightarrow \text{S}_5 + \text{SO}$	$8.00 \times 10^{-11} \exp(-300.0/T)$	(10)
R139	$\text{O} + \text{S}_7 \rightarrow \text{S}_6 + \text{SO}$	$8.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R140	$\text{O} + \text{S}_8 \rightarrow \text{S}_7 + \text{SO}$	$8.00 \times 10^{-11} \exp(-400.0/T)$	(10)
R141	$\text{O} + \text{SCl} \rightarrow \text{SO} + \text{Cl}$	1.20×10^{-10}	(9)
R142	$\text{O} + \text{SH} \rightarrow \text{H} + \text{SO}$	1.60×10^{-10}	(1)
R143	$\text{O} + \text{SO} \rightarrow \text{S} + \text{O}_2$	$6.60 \times 10^{-13} \exp(-2760.0/T)$	(3)
R144	$\text{O} + \text{SO}_2 \rightarrow \text{SO} + \text{O}_2$	$8.00 \times 10^{-12} \exp(-9800.0/T)$	(3)
R145	$\text{O} + \text{SO}_3 \rightarrow \text{SO}_2 + \text{O}_2$	$2.32 \times 10^{-16} \exp(-487.0/T)$	(10)
R146	$\text{O}_2(^1\Delta) + \text{CO} \rightarrow \text{O}_2 + \text{CO}$	1.00×10^{-20}	(3)
R147	$\text{O}_2(^1\Delta) + \text{CO}_2 \rightarrow \text{O}_2 + \text{CO}_2$	2.00×10^{-21}	(15)
R148	$\text{O}_2(^1\Delta) + \text{H}_2\text{O} \rightarrow \text{O}_2 + \text{H}_2\text{O}$	4.80×10^{-18}	(1)
R149	$\text{O}_2(^1\Delta) + \text{N} \rightarrow \text{NO} + \text{O}$	9.00×10^{-17}	(1)
R150	$\text{O}_2(^1\Delta) + \text{N}_2 \rightarrow \text{O}_2 + \text{N}_2$	1.00×10^{-20}	(1)
R151	$\text{O}_2(^1\Delta) + \text{O} \rightarrow \text{O}_2 + \text{O}$	2.00×10^{-16}	(1)
R152	$\text{O}_2(^1\Delta) + \text{O}_2 \rightarrow \text{O}_2 + \text{O}_2$	$3.60 \times 10^{-18} \exp(-220.0/T)$	(1)
R153	$\text{O}_2(^1\Delta) + \text{O}_3 \rightarrow \text{O}_2 + \text{O}_2 + \text{O}$	$5.20 \times 10^{-11} \exp(-2840.0/T)$	(1)
R154	$\text{O}_3 + \text{HNO}_2 \rightarrow \text{O}_2 + \text{HNO}_3$	5.00×10^{-19}	(1)
R155	$\text{O}_3 + \text{NO} \rightarrow \text{NO}_2 + \text{O}_2$	$3.00 \times 10^{-12} \exp(-1500.0/T)$	(1)
R156	$\text{O}_3 + \text{NO}_2 \rightarrow \text{NO}_3 + \text{O}_2$	$1.20 \times 10^{-13} \exp(-2450.0/T)$	(1)
R157	$\text{OH} + \text{CO} \rightarrow \text{CO}_2 + \text{H}$	2.40×10^{-13}	(1)
R158	$\text{OH} + \text{COCl} \rightarrow \text{HOCl} + \text{CO}$	1.50×10^{-10}	(3)
R159	$\text{OH} + \text{COS} \rightarrow \text{CO}_2 + \text{SH}$	$1.10 \times 10^{-13} \exp(-1200.0/T)$	(11)
R160	$\text{OH} + \text{Cl}_2 \rightarrow \text{Cl} + \text{HOCl}$	$2.60 \times 10^{-12} \exp(-1100.0/T)$	(1)
R161	$\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$	$2.80 \times 10^{-12} \exp(-1800.0/T)$	(1)
R162	$\text{OH} + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{HO}_2$	$2.90 \times 10^{-12} \exp(-160.0/T)$	(4)
R163	$\text{OH} + \text{H}_2\text{S} \rightarrow \text{H}_2\text{O} + \text{SH}$	$6.10 \times 10^{-12} \exp(-75.0/T)$	(11)
R164	$\text{OH} + \text{HCl} \rightarrow \text{Cl} + \text{H}_2\text{O}$	$1.80 \times 10^{-12} \exp(-250.0/T)$	(1)
R165	$\text{OH} + \text{HNO} \rightarrow \text{H}_2\text{O} + \text{NO}$	5.00×10^{-11}	(22)
R166	$\text{OH} + \text{HNO}_2 \rightarrow \text{H}_2\text{O} + \text{NO}_2$	$1.80 \times 10^{-11} \exp(-390.0/T)$	(4)
R167	$\text{OH} + \text{HNO}_3 \rightarrow \text{NO}_3 + \text{H}_2\text{O}$	$7.20 \times 10^{-15} \exp(785.0/T)$	(4)
R168	$\text{OH} + \text{HO}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$	$4.80 \times 10^{-11} \exp(250.0/T)$	(1)
R169	$\text{OH} + \text{HO}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2(^1\Delta)$	$9.60 \times 10^{-13} \exp(250.0/T)$	(4)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
R170	$\text{OH} + \text{HOCl} \rightarrow \text{H}_2\text{O} + \text{ClO}$	$3.00 \times 10^{-12} \exp(-500.0/T)$	(1)
R171	$\text{OH} + \text{NO}_3 \rightarrow \text{HO}_2 + \text{NO}_2$	2.20×10^{-11}	(1)
R172	$\text{OH} + \text{O}_3 \rightarrow \text{HO}_2 + \text{O}_2$	$1.70 \times 10^{-12} \exp(-940.0/T)$	(1)
R173	$\text{OH} + \text{O}_3 \rightarrow \text{HO}_2 + \text{O}_2(^1\Delta)$	$3.20 \times 10^{-14} \exp(-940.0/T)$	(3)
R174	$\text{OH} + \text{OH} \rightarrow \text{H}_2\text{O} + \text{O}$	1.80×10^{-12}	(1)
R175	$\text{OH} + \text{SH} \rightarrow \text{S} + \text{H}_2\text{O}$	2.50×10^{-12}	(2)
R176	$\text{OH} + \text{SO} \rightarrow \text{SO}_2 + \text{H}$	$2.60 \times 10^{-11} \exp(330.0/T)$	(1)
R177	$\text{OSCl} + \text{M} \rightarrow \text{SO} + \text{Cl} + \text{M}$	$7.21 \times 10^{-26} T^{-5.000} \exp(-5000.0/T)$	(3)
R178	$\text{O}(^1\text{D}) + \text{CO}_2 \rightarrow \text{O} + \text{CO}_2$	$7.50 \times 10^{-11} \exp(115.0/T)$	(1)
R179	$\text{O}(^1\text{D}) + \text{COCl}_2 \rightarrow \text{COCl} + \text{ClO}$	$4.40 \times 10^{-11} \exp(30.0/T)$	(3)
R180	$\text{O}(^1\text{D}) + \text{COCl}_2 \rightarrow \text{Cl}_2 + \text{CO}_2$	$1.76 \times 10^{-10} \exp(30.0/T)$	(4)
R181	$\text{O}(^1\text{D}) + \text{Cl}_2 \rightarrow \text{Cl} + \text{ClO}$	2.03×10^{-10}	(1)
R182	$\text{O}(^1\text{D}) + \text{Cl}_2 \rightarrow \text{Cl}_2 + \text{O}$	6.75×10^{-11}	(1)
R183	$\text{O}(^1\text{D}) + \text{H}_2 \rightarrow \text{H} + \text{OH}$	1.20×10^{-10}	(1)
R184	$\text{O}(^1\text{D}) + \text{H}_2\text{O} \rightarrow \text{OH} + \text{OH}$	$1.63 \times 10^{-10} \exp(60.0/T)$	(1)
R185	$\text{O}(^1\text{D}) + \text{HCl} \rightarrow \text{Cl} + \text{OH}$	1.00×10^{-10}	(1)
R186	$\text{O}(^1\text{D}) + \text{HCl} \rightarrow \text{ClO} + \text{H}$	3.30×10^{-11}	(1)
R187	$\text{O}(^1\text{D}) + \text{HCl} \rightarrow \text{O} + \text{HCl}$	1.80×10^{-11}	(1)
R188	$\text{O}(^1\text{D}) + \text{N}_2 \rightarrow \text{O} + \text{N}_2$	$2.15 \times 10^{-11} \exp(110.0/T)$	(1)
R189	$\text{O}(^1\text{D}) + \text{N}_2\text{O} \rightarrow \text{N}_2 + \text{O}_2$	$4.64 \times 10^{-11} \exp(20.0/T)$	(1)
R190	$\text{O}(^1\text{D}) + \text{N}_2\text{O} \rightarrow \text{NO} + \text{NO}$	$7.26 \times 10^{-11} \exp(20.0/T)$	(1)
R191	$\text{O}(^1\text{D}) + \text{O}_2 \rightarrow \text{O} + \text{O}_2$	$2.64 \times 10^{-11} \exp(55.0/T)$	(1)
R192	$\text{O}(^1\text{D}) + \text{O}_3 \rightarrow \text{O} + \text{O} + \text{O}_2$	1.20×10^{-10}	(1)
R193	$\text{O}(^1\text{D}) + \text{O}_3 \rightarrow \text{O}_2 + \text{O}_2$	1.20×10^{-10}	(1)
R194	$\text{O}(^1\text{D}) + \text{SO}_2 \rightarrow \text{SO} + \text{O}_2$	1.30×10^{-10}	(10)
R195	$\text{O}(^1\text{D}) + \text{SO}_2 \rightarrow \text{SO}_2 + \text{O}$	7.00×10^{-11}	(10)
R196	$\text{S} + \text{COCl} \rightarrow \text{CO} + \text{SCl}$	3.00×10^{-12}	(15)
R197	$\text{S} + \text{COCl} \rightarrow \text{COS} + \text{Cl}$	3.00×10^{-12}	(15)
R198	$\text{S} + \text{COS} \rightarrow \text{S}_2 + \text{CO}$	$2.40 \times 10^{-24} T^{4.000} \exp(-580.0/T)$	(23)
R199	$\text{S} + \text{Cl}_2 \rightarrow \text{SCl} + \text{Cl}$	$2.80 \times 10^{-11} \exp(-290.0/T)$	(9)
R200	$\text{S} + \text{ClCO}_3 \rightarrow \text{Cl} + \text{SO} + \text{CO}_2$	3.00×10^{-11}	(3)
R201	$\text{S} + \text{ClO} \rightarrow \text{SO} + \text{Cl}$	4.00×10^{-11}	(10)
R202	$\text{S} + \text{ClSO}_2 \rightarrow \text{SO}_2 + \text{SCl}$	1.00×10^{-11}	(10)
R203	$\text{S} + \text{H}_2\text{O} \rightarrow \text{OH} + \text{SH}$	$4.70 \times 10^{-11} \exp(-17700.0/T)$	(2)
R204	$\text{S} + \text{HCl} \rightarrow \text{SH} + \text{Cl}$	$1.90 \times 10^{-10} \exp(-9380.0/T)$	(2)
R205	$\text{S} + \text{HO}_2 \rightarrow \text{SO} + \text{OH}$	$3.00 \times 10^{-11} \exp(200.0/T)$	(8)
R206	$\text{S} + \text{O}_2 \rightarrow \text{SO} + \text{O}$	$1.60 \times 10^{-12} \exp(100.0/T)$	(11)
R207	$\text{S} + \text{O}_3 \rightarrow \text{SO} + \text{O}_2$	1.20×10^{-11}	(1)
R208	$\text{S} + \text{OH} \rightarrow \text{SO} + \text{H}$	6.60×10^{-11}	(1)
R209	$\text{S} + \text{OSCl} \rightarrow \text{S}_2\text{O} + \text{Cl}$	$5.00 \times 10^{-11} \exp(-600.0/T)$	(10)
R210	$\text{S} + \text{OSCl} \rightarrow \text{SO} + \text{SCl}$	$2.00 \times 10^{-11} \exp(-600.0/T)$	(10)
R211	$\text{S} + \text{S}_2\text{O} \rightarrow \text{S}_2 + \text{SO}$	$1.00 \times 10^{-12} \exp(-1200.0/T)$	(10)
R212	$\text{S} + \text{S}_3 \rightarrow \text{S}_2 + \text{S}_2$	8.00×10^{-11}	(10)
R213	$\text{S} + \text{S}_4 \rightarrow \text{S}_2 + \text{S}_3$	8.00×10^{-11}	(10)
R214	$\text{S} + \text{S}_5 \rightarrow \text{S}_2 + \text{S}_4$	$5.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R215	$\text{S} + \text{S}_5 \rightarrow \text{S}_3 + \text{S}_3$	$3.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R216	$\text{S} + \text{S}_6 \rightarrow \text{S}_2 + \text{S}_5$	$5.00 \times 10^{-11} \exp(-300.0/T)$	(10)
R217	$\text{S} + \text{S}_6 \rightarrow \text{S}_3 + \text{S}_4$	$3.00 \times 10^{-11} \exp(-300.0/T)$	(10)
R218	$\text{S} + \text{S}_7 \rightarrow \text{S}_2 + \text{S}_6$	$4.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R219	$\text{S} + \text{S}_7 \rightarrow \text{S}_3 + \text{S}_5$	$2.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R220	$\text{S} + \text{S}_7 \rightarrow \text{S}_4 + \text{S}_4$	$2.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R221	$\text{S} + \text{S}_8 \rightarrow \text{S}_2 + \text{S}_7$	$4.00 \times 10^{-11} \exp(-400.0/T)$	(10)
R222	$\text{S} + \text{S}_8 \rightarrow \text{S}_3 + \text{S}_6$	$2.00 \times 10^{-11} \exp(-400.0/T)$	(10)
R223	$\text{S} + \text{S}_8 \rightarrow \text{S}_4 + \text{S}_5$	$2.00 \times 10^{-11} \exp(-400.0/T)$	(10)
R224	$\text{S} + \text{SCl} \rightarrow \text{S}_2 + \text{Cl}$	1.00×10^{-11}	(9)
R225	$\text{S} + \text{SNO} \rightarrow \text{S}_2 + \text{NO}$	5.00×10^{-11}	(2)
R226	$\text{S} + \text{SO}_2 \rightarrow \text{SO} + \text{SO}$	$2.30 \times 10^{-11} \exp(-5200.0/T)$	(2)
R227	$\text{S} + \text{SO}_3 \rightarrow \text{SO}_2 + \text{SO}$	1.00×10^{-16}	(10)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
R228	$S_2 + CO \rightarrow S + COS$	$1.00 \times 10^{-12} \exp(-17460.0/T)$	(2)
R229	$S_2 + ClSO_2 \rightarrow SO_2 + ClS_2$	$5.00 \times 10^{-11} \exp(-800.0/T)$	(10)
R230	$S_2 + M \rightarrow S + S + M$	$k_{eq}=2.68 \times 10^{-25} \exp(50860.0/T)$	(2)
R231	$S_2 + S_2O_2 \rightarrow S_2O + S_2O$	3.30×10^{-14}	(3)
R232	$S_2 + S_8 \rightarrow S_5 + S_5$	$1.00 \times 10^{-11} \exp(-1400.0/T)$	(10)
R233	$S_2 + SCl \rightarrow S_3 + Cl$	2.00×10^{-11}	(10)
R234	$S_2 + SNO \rightarrow S_3 + NO$	1.00×10^{-17}	(2)
R235	$S_2 + SO_3 \rightarrow S_2O + SO_2$	2.00×10^{-16}	(10)
R236	$S_2O + S_2O \rightarrow S_3 + SO_2$	1.00×10^{-14}	(3)
R237	$S_2O_2 + M \rightarrow SO + SO + M$	$k_{eq}=1.00 \times 10^{-28} \exp(6000.0/T)$	(3)
R238	$S_3 + S_4 \rightarrow S_2 + S_5$	$4.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R239	$S_3 + S_5 \rightarrow S_2 + S_6$	$4.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R240	$S_3 + S_6 \rightarrow S_2 + S_7$	$4.00 \times 10^{-12} \exp(-300.0/T)$	(10)
R241	$S_3 + S_7 \rightarrow S_2 + S_8$	$3.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R242	$S_3 + S_7 \rightarrow S_4 + S_6$	$1.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R243	$S_3 + S_7 \rightarrow S_5 + S_5$	$1.00 \times 10^{-11} \exp(-200.0/T)$	(10)
R244	$S_3 + SO \rightarrow S_2O + S_2$	1.00×10^{-12}	(10)
R245	$S_4 + M \rightarrow S_2 + S_2 + M$	$k_{eq}=2.71 \times 10^{-27} \exp(13100.0/T)$	(15)
R246	$S_4 + S_5 \rightarrow S_2 + S_7$	$2.00 \times 10^{-12} \exp(-200.0/T)$	(10)
R247	$S_4 + S_5 \rightarrow S_3 + S_6$	$2.00 \times 10^{-12} \exp(-200.0/T)$	(10)
R248	$S_4 + S_6 \rightarrow S_2 + S_8$	$2.00 \times 10^{-12} \exp(-300.0/T)$	(10)
R249	$S_4 + S_6 \rightarrow S_5 + S_5$	$2.00 \times 10^{-12} \exp(-300.0/T)$	(10)
R250	$S_4 + S_7 \rightarrow S_3 + S_8$	$5.00 \times 10^{-12} \exp(-200.0/T)$	(10)
R251	$S_4 + S_7 \rightarrow S_5 + S_6$	$5.00 \times 10^{-12} \exp(-200.0/T)$	(10)
R252	$S_6 + M \rightarrow S_3 + S_3 + M$	$k_{eq}=2.41 \times 10^{-29} \exp(21483.0/T)$	(15)
R253	$S_8 + M \rightarrow S_4 + S_4 + M$	$k_{eq}=1.17 \times 10^{-29} \exp(22654.0/T)$	(15)
R254	$SCl + Cl_2 \rightarrow SCl_2 + Cl$	7.00×10^{-14}	(9)
R255	$SCl + ClS_2 \rightarrow SCl_2 + S_2$	1.00×10^{-12}	(3)
R256	$SCl + ClSO_2 \rightarrow SO_2 + SCl_2$	5.00×10^{-12}	(9)
R257	$SCl + O_2 \rightarrow SO + ClO$	2.00×10^{-15}	(9)
R258	$SCl + SCl \rightarrow ClS_2 + Cl$	5.40×10^{-11}	(9)
R259	$SCl + SCl \rightarrow S_2 + Cl_2$	6.00×10^{-12}	(9)
R260	$SCl + SCl \rightarrow SCl_2 + S$	7.50×10^{-12}	(3)
R261	$SCl + SO \rightarrow S_2O + Cl$	1.00×10^{-11}	(10)
R262	$SCl_2 + SCl_2 \rightarrow S_2Cl_2 + Cl_2$	1.00×10^{-20}	(3)
R263	$SH + CO_2 \rightarrow OH + COS$	$2.60 \times 10^{-13} \exp(-19360.0/T)$	(2)
R264	$SH + Cl_2 \rightarrow HSCl + Cl$	$1.40 \times 10^{-11} \exp(-690.0/T)$	(11)
R265	$SH + H_2 \rightarrow H_2S + H$	$3.00 \times 10^{-11} \exp(-7930.0/T)$	(2)
R266	$SH + H_2O \rightarrow H_2S + OH$	$1.00 \times 10^{-11} \exp(-14160.0/T)$	(2)
R267	$SH + H_2O_2 \rightarrow H_2S + HO_2$	5.00×10^{-15}	(1)
R268	$SH + HCl \rightarrow H_2S + Cl$	$7.60 \times 10^{-12} \exp(-5750.0/T)$	(2)
R269	$SH + O_2 \rightarrow OH + SO$	4.00×10^{-19}	(1)
R270	$SH + S \rightarrow H + S_2$	4.50×10^{-11}	(14)
R271	$SH + SCl \rightarrow S_2 + HCl$	$6.00 \times 10^{-13} \exp(230.0/T)$	(2)
R272	$SH + SH \rightarrow H_2S + S$	4.00×10^{-11}	(24)
R273	$SO + CO_2 \rightarrow SO_2 + CO$	$1.50 \times 10^{-11} \exp(-22000.0/T)$	(2)
R274	$SO + ClSO_2 \rightarrow OSCl + SO_2$	$5.00 \times 10^{-11} \exp(-800.0/T)$	(10)
R275	$SO + H \rightarrow S + OH$	$4.00 \times 10^{-10} \exp(-11200.0/T)$	(2)
R276	$SO + NO_2 \rightarrow SO_2 + NO$	1.40×10^{-11}	(1)
R277	$SO + O_2 \rightarrow SO_2 + O$	$1.60 \times 10^{-13} \exp(-2280.0/T)$	(20)
R278	$SO + O_3 \rightarrow SO_2 + O_2$	$3.40 \times 10^{-12} \exp(-1100.0/T)$	(1)
R279	$SO + O_3 \rightarrow SO_2 + O_2(^1\Delta)$	$3.60 \times 10^{-13} \exp(-1100.0/T)$	(4)
R280	$SO + OSCl \rightarrow SO_2 + SCl$	6.00×10^{-13}	(9)
R281	$SO + S_2O_2 \rightarrow S_2O + SO_2$	3.30×10^{-14}	(10)
R282	$SO + SO \rightarrow SO_2 + S$	$1.00 \times 10^{-12} \exp(-1700.0/T)$	(11)
R283	$SO + SO_3 \rightarrow SO_2 + SO_2$	2.00×10^{-15}	(10)
R284	$SO_2 + H \rightarrow SO + OH$	$3.70 \times 10^{-9} \exp(-11200.0/T)$	(2)
R285	$SO_2 + NO_2 \rightarrow SO_3 + NO$	2.00×10^{-26}	(1)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
R286	$\text{SO}_2 + \text{NO}_3 \rightarrow \text{SO}_3 + \text{NO}_2$	7.00×10^{-21}	(1)
R287	$\text{SO}_2 + \text{O}_3 \rightarrow \text{SO}_3 + \text{O}_2$	$3.00 \times 10^{-12} \exp(-7000.0/T)$	(1)
R288	$\text{SO}_2 + \text{O}_3 \rightarrow \text{SO}_3 + \text{O}_2(^1\Delta)$	$6.00 \times 10^{-14} \exp(-7000.0/T)$	(15)
R289	$\text{SO}_2\text{Cl}_2 + \text{H} \rightarrow \text{ClSO}_2 + \text{HCl}$	1.00×10^{-11}	(2)
R290	$\text{SO}_2\text{Cl}_2 + \text{S} \rightarrow \text{ClSO}_2 + \text{SCl}$	1.00×10^{-12}	(2)
R291	$\text{SO}_2\text{Cl}_2 + \text{SO}_2 \rightarrow \text{ClSO}_2 + \text{ClSO}_2$	$1.00 \times 10^{-12} \exp(-11000.0/T)$	(2)
R292	$\text{SO}_3 + \text{CO} \rightarrow \text{CO}_2 + \text{SO}_2$	$1.00 \times 10^{-11} \exp(-13000.0/T)$	(25)
R293*	$\text{SO}_2 + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{SO}_2(\text{l}) + \text{H}_2\text{SO}_4(\text{l})$	$1.00 \times 10^{-32} \exp(9000.0/T)$	(26)
R294*	$\text{SO}_2(\text{l}) + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{SO}_2 + \text{H}_2\text{SO}_4(\text{l})$	$6.67 \times 10^{-36} \exp(9000.0/T)$	(26)
R295*	$\text{SO}_2(\text{l}) + \text{H}_2\text{SO}_4 \rightarrow \text{SO}_2 + \text{H}_2\text{SO}_4$	$2.20 \times 10^{-04} \exp(-10000.0/T)$	(26)
R296	$\text{COCl} + \text{O}_2 + \text{M} \rightarrow \text{ClCO}_3 + \text{M}$	$k_0 = 5.70 \times 10^{-15} \exp(500.0/T)/$ $(1 \times 10^7 + 0.05 n_{\text{atm}})$	(8)
R297	$\text{COS} + \text{SO}_3 + \text{SO}_3 \rightarrow \text{SO}_2 + \text{SO}_2 + \text{SO}_2 + \text{CO}$	$k_0 = 3.00 \times 10^{-32}$	(4)
R298	$\text{Cl} + \text{CO} + \text{CO}_2 \rightarrow \text{COCl} + \text{CO}_2$	$k_0 = 3.37 \times 10^{-24} T^{-3.800}$	(3)
R299	$\text{Cl} + \text{CO} + \text{N}_2 \rightarrow \text{COCl} + \text{N}_2$	$k_0 = 3.37 \times 10^{-24} T^{-3.800}$	(3)
R300	$\text{Cl} + \text{Cl} + \text{CO}_2 \rightarrow \text{Cl}_2 + \text{CO}_2$	$k_0 = 2.60 \times 10^{-33} \exp(900.0/T)$	(3)
R301	$\text{Cl} + \text{Cl} + \text{N}_2 \rightarrow \text{Cl}_2 + \text{N}_2$	$k_0 = 6.10 \times 10^{-34} \exp(900.0/T)$	(3)
R302	$\text{Cl} + \text{ClSO}_2 + \text{M} \rightarrow \text{SO}_2\text{Cl}_2 + \text{M}$	$k_0 = 1.00 \times 10^{-23} T^{-2.000}$	(10)
R303	$\text{Cl} + \text{H} + \text{M} \rightarrow \text{HCl} + \text{M}$	$k_0 = 1.00 \times 10^{-32}$	(8)
R304	$\text{Cl} + \text{O} + \text{M} \rightarrow \text{ClO} + \text{M}$	$k_0 = 5.00 \times 10^{-32}$	(8)
R305	$\text{Cl} + \text{SCl} + \text{M} \rightarrow \text{SCl}_2 + \text{M}$	$k_0 = 1.00 \times 10^{-30}$ $k_\infty = 5.00 \times 10^{-11}$	(9)
R306	$\text{Cl} + \text{SO} + \text{M} \rightarrow \text{OSCl} + \text{M}$	$k_0 = 7.30 \times 10^{-21} T^{-5.000}$	(9)
R307	$\text{Cl} + \text{SO}_2 + \text{M} \rightarrow \text{ClSO}_2 + \text{M}$	$k_0 = 1.30 \times 10^{-34} \exp(940.0/T)$	(3)
R308	$\text{ClO} + \text{SO}_2 + \text{M} \rightarrow \text{Cl} + \text{SO}_3 + \text{M}$	$k_0 = 1.00 \times 10^{-35}$ $k_\infty = 4.00 \times 10^{-19}$	(3)
R309	$\text{H} + \text{H} + \text{M} \rightarrow \text{H}_2 + \text{M}$	$k_0 = 2.70 \times 10^{-31} T^{-0.600}$	(21)
R310	$\text{H} + \text{NO} + \text{M} \rightarrow \text{HNO} + \text{M}$	$k_0 = 3.23 \times 10^{-32}$	(27)
R311	$\text{H} + \text{O}_2 + \text{CO}_2 \rightarrow \text{HO}_2 + \text{CO}_2$	$k_0 = 2.41 \times 10^{-28} T^{-1.300}$ $k_\infty = 2.48 \times 10^{-10} T^{0.200} \exp(300.0/T)$	(3)
R312	$\text{H} + \text{O}_2 + \text{N}_2 \rightarrow \text{HO}_2 + \text{N}_2$	$k_0 = 7.31 \times 10^{-29} T^{-1.300}$ $k_\infty = 2.40 \times 10^{-11} T^{0.200}$	(4)
R313	$\text{H} + \text{OH} + \text{CO}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$	$k_0 = 7.70 \times 10^{-26} T^{-2.000}$	(21)
R314	$\text{H} + \text{OH} + \text{N}_2 \rightarrow \text{H}_2\text{O} + \text{N}_2$	$k_0 = 6.10 \times 10^{-26} T^{-2.000}$	(21)
R315	$\text{H} + \text{SH} + \text{M} \rightarrow \text{H}_2\text{S} + \text{M}$	$k_0 = 9.00 \times 10^{-26} T^{-2.000}$	(2)
R316	$\text{HO}_2 + \text{HO}_2 + \text{M} \rightarrow \text{H}_2\text{O}_2 + \text{O}_2 + \text{M}$	$k_0 = 2.10 \times 10^{-33} \exp(920.0/T)$	(1)
R317	$\text{N} + \text{N} + \text{M} \rightarrow \text{N}_2 + \text{M}$	$k_0 = 8.27 \times 10^{-34} \exp(490.0/T)$	(28)
R318	$\text{O} + \text{CO} + \text{CO} \rightarrow \text{CO}_2 + \text{CO}$	$k_0 = 6.50 \times 10^{-33} \exp(-2180.0/T)$	(3)
R319	$\text{O} + \text{CO} + \text{M} \rightarrow \text{CO}_2 + \text{M}$	$k_0 = 1.70 \times 10^{-33} \exp(-1510.0/T)$ $k_\infty = 2.66 \times 10^{-14} \exp(-1459.0/T)$	(29)
R320	$\text{O} + \text{H} + \text{M} \rightarrow \text{OH} + \text{M}$	$k_0 = 1.30 \times 10^{-29} T^{-1.000}$	(29)
R321	$\text{O} + \text{NO} + \text{M} \rightarrow \text{NO}_2 + \text{M}$	$k_0 = 4.68 \times 10^{-28} T^{-1.500}$ $k_\infty = 3.00 \times 10^{-11}$	(1)
R322	$\text{O} + \text{NO}_2 + \text{M} \rightarrow \text{NO}_3 + \text{M}$	$k_0 = 3.09 \times 10^{-27} T^{-1.600}$ $k_\infty = 7.19 \times 10^{-11} T^{-0.200}$	(1)
R323	$\text{O} + \text{O} + \text{CO} \rightarrow \text{CO}_2 + \text{O}$	$k_0 = 3.40 \times 10^{-33} \exp(-2180.0/T)$	(3)
R324	$\text{O} + \text{O} + \text{CO}_2 \rightarrow \text{O}_2 + \text{CO}_2$	$k_0 = 5.20 \times 10^{-28} T^{-2.000}$	(15)
R325	$\text{O} + \text{O} + \text{CO}_2 \rightarrow \text{O}_2(^1\Delta) + \text{CO}_2$	$k_0 = 7.70 \times 10^{-28} T^{-2.000}$	(15)
R326	$\text{O} + \text{O} + \text{O}_2 \rightarrow \text{O}_3 + \text{O}$	$k_0 = 5.20 \times 10^{-28} T^{-2.400}$ $k_\infty = 2.80 \times 10^{-12}$	(3)
R327	$\text{O} + \text{O}_2 + \text{CO} \rightarrow \text{O}_3 + \text{CO}$	$k_0 = 1.04 \times 10^{-27} T^{-2.500}$ $k_\infty = 2.80 \times 10^{-12}$	(3)
R328	$\text{O} + \text{O}_2 + \text{CO}_2 \rightarrow \text{O}_3 + \text{CO}_2$	$k_0 = 2.18 \times 10^{-27} T^{-2.500}$ $k_\infty = 2.80 \times 10^{-12}$	(3)
R329	$\text{O} + \text{O}_2 + \text{CO}_2 \rightarrow \text{O}_3 + \text{CO}_2$	$k_0 = 5.29 \times 10^{-28} T^{-2.400}$	(1)
R330	$\text{O} + \text{O}_2 + \text{N}_2 \rightarrow \text{O}_3 + \text{N}_2$	$k_0 = 8.73 \times 10^{-28} T^{-2.500}$ $k_\infty = 2.80 \times 10^{-12}$	(3)
R331	$\text{O} + \text{O}_2 + \text{O}_2 \rightarrow \text{O}_3 + \text{O}_2$	$k_0 = 5.20 \times 10^{-28} T^{-2.400}$	(3)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
R332	$\text{O} + \text{SO} + \text{M} \rightarrow \text{SO}_2 + \text{M}$	$k_\infty = 2.80 \times 10^{-12}$ $k_0 = 4.18 \times 10^{-30}$	(3)
R333	$\text{O} + \text{SO}_2 + \text{M} \rightarrow \text{SO}_3 + \text{M}$	$k_\infty = 4.35 \times 10^{-10}$ $k_0 = 6.56 \times 10^{-38} T^{2.000}$	(20)
R334	$\text{OH} + \text{NO} + \text{M} \rightarrow \text{HNO}_2 + \text{M}$	$k_\infty = 4.83 \times 10^{-18} T^{1.800}$ $k_0 = 1.93 \times 10^{-24} T^{-2.600}$	(1)
R335	$\text{OH} + \text{NO}_2 + \text{M} \rightarrow \text{HNO}_3 + \text{M}$	$k_\infty = 6.37 \times 10^{-11} T^{-0.100}$ $k_0 = 4.15 \times 10^{-22} T^{-3.900}$	(1)
R336	$\text{OH} + \text{OH} + \text{M} \rightarrow \text{H}_2\text{O}_2 + \text{M}$	$k_\infty = 7.25 \times 10^{-10} T^{-0.500}$ $k_0 = 2.06 \times 10^{-28} T^{-1.000}$	(1)
R337	$\text{OH} + \text{SO}_2 + \text{M} \rightarrow \text{HSO}_3 + \text{M}$	$k_\infty = 2.60 \times 10^{-11}$ $k_0 = 4.04 \times 10^{-21} T^{-4.100}$	(1)
R338	$\text{O}(^1\text{D}) + \text{N}_2 + \text{M} \rightarrow \text{N}_2\text{O} + \text{M}$	$k_\infty = 5.31 \times 10^{-12} T^{-0.200}$ $k_0 = 4.72 \times 10^{-34} T^{-0.900}$	(1)
R339	$\text{S} + \text{CO} + \text{M} \rightarrow \text{COS} + \text{M}$	$k_0 = 4.00 \times 10^{-33} \exp(-1940.0/T)$	(15)
R340	$\text{S} + \text{Cl} + \text{M} \rightarrow \text{SCl} + \text{M}$	$k_0 = 1.00 \times 10^{-29} T^{-1.000}$	(10)
R341	$\text{S} + \text{NO} + \text{M} \rightarrow \text{SNO} + \text{M}$	$k_0 = 3.00 \times 10^{-32} \exp(940.0/T)$	(30)
R342	$\text{S} + \text{O} + \text{M} \rightarrow \text{SO} + \text{M}$	$k_0 = 1.50 \times 10^{-34} \exp(900.0/T)$	(10)
R343	$\text{S} + \text{S} + \text{M} \rightarrow \text{S}_2 + \text{M}$	$k_0 = 1.18 \times 10^{-29}$ $k_\infty = 1.00 \times 10^{-10}$	(3)
R344	$\text{S} + \text{S}_2 + \text{M} \rightarrow \text{S}_3 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(10)
R345	$\text{S} + \text{S}_3 + \text{M} \rightarrow \text{S}_4 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(10)
R346	$\text{S} + \text{S}_4 + \text{M} \rightarrow \text{S}_5 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(10)
R347	$\text{S} + \text{S}_5 + \text{M} \rightarrow \text{S}_6 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R348	$\text{S} + \text{S}_6 + \text{M} \rightarrow \text{S}_7 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R349	$\text{S} + \text{S}_7 + \text{M} \rightarrow \text{S}_8 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R350	$\text{S} + \text{SO} + \text{M} \rightarrow \text{S}_2\text{O} + \text{M}$	$k_0 = 3.30 \times 10^{-26} T^{-2.000}$	(10)
R351	$\text{S}_2 + \text{S}_2 + \text{M} \rightarrow \text{S}_4 + \text{M}$	$k_0 = 2.20 \times 10^{-29}$ $k_\infty = 1.00 \times 10^{-10}$	(10)
R352	$\text{S}_2 + \text{S}_3 + \text{M} \rightarrow \text{S}_5 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R353	$\text{S}_2 + \text{S}_4 + \text{M} \rightarrow \text{S}_6 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(10)
R354	$\text{S}_2 + \text{S}_5 + \text{M} \rightarrow \text{S}_7 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R355	$\text{S}_2 + \text{S}_6 + \text{M} \rightarrow \text{S}_8 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R356	$\text{S}_3 + \text{S}_3 + \text{M} \rightarrow \text{S}_6 + \text{M}$	$k_0 = 1.00 \times 10^{-30}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R357	$\text{S}_3 + \text{S}_4 + \text{M} \rightarrow \text{S}_7 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(10)
R358	$\text{S}_3 + \text{S}_5 + \text{M} \rightarrow \text{S}_8 + \text{M}$	$k_0 = 1.00 \times 10^{-25} T^{-2.000}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R359	$\text{S}_4 + \text{S}_4 + \text{M} \rightarrow \text{S}_8 + \text{M}$	$k_0 = 1.00 \times 10^{-30}$ $k_\infty = 3.00 \times 10^{-11}$	(3)
R360	$\text{SCl} + \text{SCl} + \text{M} \rightarrow \text{S}_2\text{Cl}_2 + \text{M}$	$k_0 = 4.00 \times 10^{-31}$ $k_\infty = 4.00 \times 10^{-12}$	(3)
R361	$\text{SO} + \text{SO} + \text{M} \rightarrow \text{S}_2\text{O}_2 + \text{M}$	$k_0 = 4.40 \times 10^{-31}$ $k_\infty = 1.00 \times 10^{-11}$	(3)
R362	$\text{SO}_3 + \text{H}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$	$k_0 = 2.30 \times 10^{-43} T^{1.000} \exp(6540.0/T)$	(31)

Table S.1. continued.

Index	Reaction	Rate Coefficient	Reference
Notes. T is the temperature, n_{atm} is the atmospheric number density, k_{eq} is the equilibrium constant. The reactions (R293-295) marked by superscript ‘★’ refer to the SO_2 dissolution, which is included in the nominal model and Model A. References. (1) Burkholder et al. (2020); (2) Krasnopolsky (2007); (3) Mills (1998); (4) Bierson & Zhang (2020); (5) Krasnopolsky (2012); (6) Krasnopolsky (2013); (7) Baulch et al. (1981); (8) Yung & Demore (1982); (9) Mills & Allen (2007); (10) Moses et al. (2002); (11) Sander et al. (2006); (12) Wang et al. (2003); (13) Clyne et al. (1984); (14) Woodall et al. (2007); (15) Zhang et al. (2012); (16) He et al. (1993); (17) Bryukov et al. (1993); (18) Brune et al. (1983); (19) Atkinson et al. (1989); (20) Atkinson et al. (2004); (21) Baulch et al. (1992); (22) Sun et al. (2001); (23) Lu et al. (2006); (24) Schofield (1973); (25) Krasnopolsky & Pollack (1994); (26) Rimmer et al. (2021); (27) Riley et al. (2003); (28) Campbell & Thrush (1967); (29) Moses et al. (2000); (30) Goumri et al. (2004); (31) Lovejoy et al. (1996)			

2. Supplemental Tables: Photolysis Reactions

The photolysis reactions used in our models are collected in Table S.2:

Table S.2. Photolysis reactions

Index	Reaction	Branch Index
P363	$\text{O}_2 + h\nu \rightarrow \text{O} + \text{O}$	1
P364	$\text{O}_2 + h\nu \rightarrow \text{O} + \text{O}(^1\text{D})$	2
P365	$\text{O}_3 + h\nu \rightarrow \text{O}_2 + \text{O}$	1
P366	$\text{O}_3 + h\nu \rightarrow \text{O}_2(^1\Delta) + \text{O}(^1\text{D})$	2
P367	$\text{O}_3 + h\nu \rightarrow \text{O} + \text{O} + \text{O}$	3
P368	$\text{OH} + h\nu \rightarrow \text{O} + \text{H}$	1
P369	$\text{HO}_2 + h\nu \rightarrow \text{OH} + \text{O}$	1
P370	$\text{H}_2\text{O} + h\nu \rightarrow \text{H} + \text{OH}$	1
P371	$\text{H}_2\text{O} + h\nu \rightarrow \text{H}_2 + \text{O}(^1\text{D})$	2
P372	$\text{H}_2\text{O} + h\nu \rightarrow \text{H} + \text{H} + \text{O}$	3
P373	$\text{H}_2\text{O}_2 + h\nu \rightarrow \text{OH} + \text{OH}$	1
P374	$\text{Cl}_2 + h\nu \rightarrow \text{Cl} + \text{Cl}$	1
P375	$\text{ClO} + h\nu \rightarrow \text{Cl} + \text{O}$	1
P376	$\text{HCl} + h\nu \rightarrow \text{H} + \text{Cl}$	1
P377	$\text{HOCl} + h\nu \rightarrow \text{OH} + \text{Cl}$	1
P378	$\text{COCl}_2 + h\nu \rightarrow \text{Cl} + \text{Cl} + \text{CO}$	1
P379	$\text{CO}_2 + h\nu \rightarrow \text{CO} + \text{O}$	1
P380	$\text{CO}_2 + h\nu \rightarrow \text{CO} + \text{O}(^1\text{D})$	2
P381	$\text{S}_2 + h\nu \rightarrow \text{S} + \text{S}$	1
P382	$\text{S}_3 + h\nu \rightarrow \text{S}_2 + \text{S}$	1
P383	$\text{S}_4 + h\nu \rightarrow \text{S}_2 + \text{S}_2$	1
P384	$\text{SCl} + h\nu \rightarrow \text{S} + \text{Cl}$	1
P385	$\text{SCl}_2 + h\nu \rightarrow \text{SCl} + \text{Cl}$	1
P386	$\text{SCl}_2 + h\nu \rightarrow \text{S} + \text{Cl} + \text{Cl}$	2
P387	$\text{ClS}_2 + h\nu \rightarrow \text{S}_2 + \text{Cl}$	1
P388	$\text{SO} + h\nu \rightarrow \text{S} + \text{O}$	1
P389	$\text{SO}_2 + h\nu \rightarrow \text{SO} + \text{O}$	1
P390	$\text{SO}_2 + h\nu \rightarrow \text{S} + \text{O}_2$	2
P391	$\text{SO}_2 + h\nu \rightarrow \text{S} + \text{O} + \text{O}$	3
P392	$\text{SO}_3 + h\nu \rightarrow \text{SO}_2 + \text{O}$	1
P393	$\text{COS} + h\nu \rightarrow \text{CO} + \text{S}$	1
P394	$\text{H}_2\text{SO}_4 + h\nu \rightarrow \text{SO}_3 + \text{H}_2\text{O}$	1
P395	$\text{S}_2\text{O} + h\nu \rightarrow \text{SO} + \text{S}$	1
P396	$\text{ClCO}_3 + h\nu \rightarrow \text{CO}_2 + \text{ClO}$	1
P397	$\text{NO} + h\nu \rightarrow \text{N} + \text{O}$	1
P398	$\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$	1
P399	$\text{NO}_3 + h\nu \rightarrow \text{NO}_2 + \text{O}$	1
P400	$\text{NO}_3 + h\nu \rightarrow \text{NO} + \text{O}_2$	2
P401	$\text{N}_2 + h\nu \rightarrow \text{N} + \text{N}$	1
P402	$\text{N}_2\text{O} + h\nu \rightarrow \text{N}_2 + \text{O}(^1\text{D})$	1
P403	$\text{HNO}_2 + h\nu \rightarrow \text{OH} + \text{NO}$	1

Table S.2. continued.

Index	Reaction	Branch Index
P404	$\text{HNO}_3 + h\nu \rightarrow \text{NO}_2 + \text{OH}$	1
P405	$\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$	1
P406	$\text{S}_2\text{O}_2 + h\nu \rightarrow \text{SO} + \text{SO}$	1
P407	$\text{S}_2\text{Cl}_2 + h\nu \rightarrow \text{SCl} + \text{SCl}$	1
P408	$\text{S}_2\text{Cl}_2 + h\nu \rightarrow \text{S}_2 + \text{Cl} + \text{Cl}$	2

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